

Symmetry, Coherence, and Virtual-Pathway Organization of Soft-Space Hotspots in Low-Qubit Hilbert Spaces

Phase 3.2: A Detailed Mechanistic Study

Tom Stevns

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Abstract

Phase 3.2 develops a mechanistic explanation for the organization of “hotspots” previously observed in low-qubit Hilbert-space models. The starting point is a family of finite-dimensional Hamiltonians equipped with two perturbation classes, dephasing-type and transverse-type, and a REAL-versus-NULl comparison framework. Earlier phases established that hotspot structure is not captured by a simple coordinate prediction alone, but is nevertheless robustly detectable through perturbative and projector-based quantities.

The central advance of Phase 3.2 is the identification of a two-layer mechanism. First, an exact chiral-like symmetry provides a global $\pm E$ spectral backbone. Second, local hotspot strength is organized by coherent virtual transitions through the complementary Q -space. These virtual pathways are naturally described by degeneracy-closed operators T_G , whose sum defines an effective self-energy-like operator Σ_E . A dimensionless coherence factor

$$C_E = \frac{\|\Sigma_E\|_F}{\sum_G \|T_G\|_F}$$

was found to be the most robust local descriptor. It obeys the exact decomposition

$$C_E^2 = S_E + A_E,$$

where S_E is a pathway-concentration term and A_E is a normalized interference term. The same quantity also admits a direct effective-response interpretation:

$$\left\| \Delta H_{\text{eff}}^{(2)}(\lambda) \right\|_F = \lambda^2 C_E L_{1,E}, \quad L_{1,E} = \sum_G \|T_G\|_F.$$

A second symmetry layer was subsequently uncovered for the transverse perturbation family. Two Pauli-string operators were found that commute with the reference Hamiltonian while anticommute with the transverse perturbation. This provides a same-target $\lambda \mapsto -\lambda$ symmetry and explains the numerically observed even-order transverse response. Together, the results support a coherent mechanism in which global spectral symmetry provides the backbone, while degeneracy-closed virtual-path interference determines local hotspot organization and family-dependent response structure.

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1 Motivation and scope

The purpose of Phase 3.2 was not to search for additional empirical correlates indefinitely, but to move from robust hotspot detection toward a mechanistic explanation.

The guiding question was:

Why do certain spectral subspaces behave as strong hotspots, while nearby or symmetry-related subspaces remain comparatively weak?

Earlier work had already shown that perturbation-based quantities could identify hotspot structure even when simple coordinate-based hotspot predictions were imperfect. Phase 3.2 therefore shifted the emphasis from *where* hotspots occur to *why* some eigenspaces respond more strongly than others.

The strategy adopted in this phase was deliberately conservative:

1. freeze the previously chosen 8-qubit, 11-term reference model;
2. retain the REAL-versus-NULl comparison logic;
3. work with complete degeneracy-closed eigenspaces rather than individual basis vectors;
4. avoid uncontrolled feature proliferation;
5. demand basis-invariant operator quantities whenever possible;
6. require any proposed mechanism to survive cross-seed and finite-size tests before assigning physical significance to it.

The result is a hierarchy of statements with different logical status:

- exact algebraic identities;
- operator symmetries validated to numerical precision;
- finite-size numerical robustness;
- statistical relations to hotspot prominence.

This distinction is essential. Phase 3.2 does *not* claim an exact closed-form formula for hotspot prominence. It does, however, identify an operator structure that explains why the empirically strongest hotspot descriptor has a direct perturbative meaning.

2 Frozen model and notation

The principal reference calculations use an 8-qubit Hilbert space,

$$\dim \mathcal{H} = 2^8 = 256,$$

with a Hamiltonian constructed from 11 random Pauli terms. The frozen seed set is

$$25042000, \dots, 25042011.$$

Two perturbation families are retained throughout:

$$\text{dephasing : } Z, ZZ \text{ type terms,} \quad (1)$$

$$\text{transverse : } X, XX \text{ type terms.} \quad (2)$$

For a chosen eigenspace P of the unperturbed Hamiltonian H_0 , define

$$Q = I - P.$$

The Q -space is decomposed into complete degeneracy-closed eigenspaces P_G :

$$Q = \sum_G P_G.$$

The use of complete degeneracy-closed projectors is critical. Individual basis states inside a degenerate eigenspace are gauge dependent; the projectors P_G are not.

For target energy E and a regularized energy kernel

$$g_\eta(x) = \frac{x}{x^2 + \eta^2},$$

define the grouped virtual-pathway operators

$$T_G = g_\eta(E - E_G) P V P_G V P. \quad (3)$$

The corresponding effective second-order operator is

$$\Sigma_E = \sum_G T_G. \quad (4)$$

Equation (3) is the basic microscopic building block of the Phase 3.2 mechanism.

3 Global spectral backbone: exact $\pm E$ symmetry

A major early result of Phase 3.2 was the identification of a Pauli-string operator

$$\Gamma = \text{XIXXIIXX}$$

for the frozen reference model, satisfying

$$\Gamma H_0 \Gamma^\dagger = -H_0. \quad (5)$$

This relation implies spectral pairing: if E is an eigenvalue, then $-E$ is also an eigenvalue. At the projector level,

$$P_{-E} = \Gamma P_E \Gamma^\dagger, \quad Q_{-E} = \Gamma Q_E \Gamma^\dagger. \quad (6)$$

The perturbation families transform differently:

$$\Gamma V_D \Gamma^\dagger = -V_D, \quad (7)$$

$$\Gamma V_T \Gamma^\dagger = +V_T, \quad (8)$$

where V_D and V_T denote the dephasing and transverse perturbations.

The operator-level tests showed that the second-order effective object transforms according to

$$\Sigma_{-E} = -\Gamma \Sigma_E \Gamma^\dagger, \quad (9)$$

with residuals near machine precision in the reference calculation.

This established an important conceptual separation:

The $\pm E$ symmetry is a global algebraic backbone, but it does not by itself determine which paired eigenspaces become strong local hotspots.

This distinction motivated the subsequent search for a local mechanism.

4 From raw coupling to effective virtual-path structure

The first local mechanism probes compared several quantities:

- direct PVP structure;
- PVQ coupling;
- $\|PVQ\|_F^2$;
- energy-gap-weighted couplings;
- second-order self-energy-like quantities.

The raw P - Q coupling norm was not sufficient. In particular, the transverse family displayed cases in which $\|PVQ\|_F^2$ was nearly constant across candidate subspaces while hotspot prominence varied substantially.

This indicated that the relevant information was not simply *how strongly* P couples to Q , but *how the many virtual channels through Q combine*.

This observation led directly to the grouped decomposition

$$\Sigma_E = \sum_G T_G.$$

The conceptual change was substantial:

The hotspot mechanism is not governed by a single transition amplitude, but by the coherent vector-like sum of many operator-valued virtual pathways.

5 Interference and coherence of virtual pathways

The squared Frobenius norm of the total effective operator is

$$\|\Sigma_E\|_F^2 = \left\| \sum_G T_G \right\|_F^2 \quad (10)$$

$$= \sum_G \|T_G\|_F^2 + 2 \sum_{G < K} \text{Re Tr}(T_G^\dagger T_K). \quad (11)$$

Equation (11) separates two physically distinct effects:

- (i) the individual strength of each virtual pathway;
- (ii) cross-path interference between different pathways.

A sign-randomization test was then used as a null model. Independent sign flips of the grouped contributions preserve individual pathway norms while destroying coherent alignment. The observed REAL data showed systematic excess coherence relative to these randomized controls.

A strict confirmation stage used 1000 sign randomizations per pair and 2000 seed-level Spearman permutations. Both perturbation families passed the predeclared strong confirmation gates.

The results support the statement:

Hotspot organization is associated with coherent or cancelling alignment among virtual pathways, not merely with the magnitude of the individual pathways.

6 The coherence factor

The compact dimensionless coherence measure is

$$C_E = \frac{\|\sum_G T_G\|_F}{\sum_G \|T_G\|_F} = \frac{\|\Sigma_E\|_F}{L_{1,E}}, \quad L_{1,E} = \sum_G \|T_G\|_F. \quad (12)$$

Its interpretation is immediate:

- C_E is large when the pathway operators reinforce one another;
- C_E is small when substantial cancellation occurs;
- C_E is normalized by the total pathway budget $L_{1,E}$ and is therefore not simply a measure of total coupling strength.

Across the frozen 8-qubit seed ensemble, C_E showed a particularly strong and consistent positive relation with hotspot prominence, especially for the transverse family.

For the 8-qubit, 11-term reference ensemble, the cross-seed median Spearman correlations were approximately

$$\rho_{\text{med}}^{(D)}(C_E, \text{prominence}) \approx 0.2500, \quad (13)$$

$$\rho_{\text{med}}^{(T)}(C_E, \text{prominence}) \approx 0.5194, \quad (14)$$

with positive association in all 12 seeds for both families in the v32.21 decomposition analysis.

7 Exact coherence decomposition

Phase 3.2 then decomposed C_E into a concentration term and an interference term.

Define

$$S_E = \frac{\sum_G \|T_G\|_F^2}{(\sum_G \|T_G\|_F)^2}, \quad (15)$$

and

$$A_E = \frac{2 \sum_{G < K} \text{Re Tr}(T_G^\dagger T_K)}{(\sum_G \|T_G\|_F)^2}. \quad (16)$$

Then, directly from [Eq. \(11\)](#),

$$\boxed{C_E^2 = S_E + A_E}. \quad (17)$$

This is an exact algebraic identity, not a fitted relation.

The terms have distinct interpretations:

$$S_E : \text{pathway concentration}, \quad (18)$$

$$A_E : \text{normalized constructive/destructive interference}. \quad (19)$$

Numerically, the identity was verified at the level of

$$\max |C_E^2 - (S_E + A_E)| \sim 10^{-16}.$$

For the 8-qubit ensemble:

Table 1: Cross-seed median Spearman correlations with hotspot prominence in the v32.21 coherence decomposition.

Family	S_E	A_E	C_E
Dephasing	+0.1176	+0.2605	+0.2500
Transverse	+0.1632	+0.2946	+0.5194

The most important interpretation is not that S_E and A_E contribute equally. Rather, interference is the stronger isolated component in most cases, while the full normalized coherence factor provides the most robust combined descriptor.

8 Finite-size robustness from 7Q to 12Q

The degeneracy-closed mechanism was tested over multiple Hamiltonian term counts and multiple system sizes.

For the 11-term model, the principal 7Q–12Q results for A_E and C_E were:

These data support a finite-size robustness statement:

Table 2: Finite-size robustness of the degeneracy-closed interference and coherence descriptors. Entries are median seedwise Spearman correlations; parentheses give the number of positive seeds out of 12.

Qubits	Dephasing		Transverse	
	A_E	C_E	A_E	C_E
7Q	0.4571 (11/12)	0.4353 (11/12)	0.2519 (9/12)	0.4307 (9/12)
8Q	0.2605 (10/12)	0.2500 (12/12)	0.2843 (10/12)	0.5194 (12/12)
9Q	0.3859 (10/12)	0.4412 (11/12)	0.2505 (11/12)	0.3179 (11/12)
10Q	0.3576 (10/12)	0.5138 (12/12)	0.4044 (11/12)	0.4361 (11/12)
11Q	0.2466 (9/12)	0.3469 (10/12)	0.2905 (9/12)	0.3647 (11/12)
12Q	0.4690 (9/12)	0.3821 (11/12)	0.3883 (12/12)	0.4250 (10/12)

The virtual-path interference/coherence mechanism remains positively associated with hotspot prominence throughout the investigated 7Q–12Q range.

They do *not* establish an asymptotic theorem in the large- Q limit.

The 7Q and 12Q endpoints were revisited explicitly in the coherence decomposition framework. The same qualitative picture survived: interference remained the stronger single component, while the full coherence factor remained robust.

9 Failed and partially successful mechanism hypotheses

A scientifically important aspect of Phase 3.2 is that several plausible simplifications were tested and rejected.

9.1 Raw coupling law

A direct law based on $\|PVQ\|_F^2$ was insufficient. Hotspots could vary substantially while raw P – Q coupling remained nearly constant.

9.2 Single scalar self-energy law

Static scalar quantities built only from Σ_E produced positive signals but did not provide a universal hotspot law across both families and all seeds.

9.3 Multivariate ridge models

LOSO ridge regression gave modest positive correlations but weak explanatory power. This route was not pursued further because the aim was mechanism, not predictive feature fitting.

9.4 Q -spectrum descriptors

Additional descriptors of the Q -spectrum did not materially improve the model.

9.5 Static operator-shape descriptors

Simple static shape measures of Σ_E were not sufficient.

9.6 Perturbative susceptibility as a universal hotspot law

First- and second-order response derivatives were tested. A transverse second-order quantity showed a real cross-seed signal, but it was weaker than the static coherence factor and did not replace it as a universal law.

These negative results were decisive in redirecting the analysis toward operator-valued interference and exact symmetry rather than continued feature construction.

10 Effective-response interpretation of the coherence factor

The key analytical bridge was obtained from the regularized Feshbach-type effective operator.

For a scaled perturbation

$$H(\lambda) = H_0 + \lambda V,$$

the leading second-order effective correction in the target P -space is

$$\Delta H_{\text{eff}}^{(2)}(\lambda) = \lambda^2 \Sigma_E. \quad (20)$$

Taking the Frobenius norm and using [Eq. \(12\)](#),

$$\left\| \Delta H_{\text{eff}}^{(2)}(\lambda) \right\|_F = \lambda^2 C_E L_{1,E}. \quad (21)$$

Equation (21) is one of the central results of Phase 3.2.

It gives C_E a direct interpretation:

At fixed total virtual-pathway budget $L_{1,E}$, the coherence factor C_E is exactly the multiplicative factor controlling the norm of the leading second-order effective P -space response.

The regularized finite- λ response was also tested directly:

$$F_E(\lambda) = \lambda^2 \text{Herm} \left[B^\dagger (EI - QH_0Q - \lambda QVQ + i\eta I)^{-1} B \right], \quad (22)$$

with

$$B = QVP.$$

The grouped representation and the direct resolvent representation of Σ_E agreed to residuals of order 10^{-12} or smaller in the reference calculations.

11 Perturbative order structure

Expanding the resolvent gives

$$F_E(\lambda) = \lambda^2 K_2 + \lambda^3 K_3 + \lambda^4 K_4 + \lambda^5 K_5 + \cdots, \quad (23)$$

where

$$K_2 = \text{Herm} \left[B^\dagger R_0 B \right], \quad (24)$$

$$K_3 = \text{Herm} \left[B^\dagger R_0 W R_0 B \right], \quad (25)$$

$$K_4 = \text{Herm} \left[B^\dagger R_0 W R_0 W R_0 B \right], \quad (26)$$

with

$$R_0 = (EI - QH_0Q + i\eta I)^{-1}, \quad W = QVQ.$$

The dephasing and transverse families behaved very differently.

For dephasing:

$$\text{median} \frac{\|K_3\|_F}{\|K_2\|_F} \approx 7.7 \times 10^{-2},$$

so the cubic term is genuinely present.

For transverse:

$$\text{median} \frac{\|K_3\|_F}{\|K_2\|_F} \approx 2.0 \times 10^{-12},$$

with a maximum across groups of only approximately

$$4.4 \times 10^{-11}.$$

The fifth-order coefficient was likewise strongly suppressed:

$$\text{median} \frac{\|K_5\|_F}{\|K_2\|_F} \approx 5.2 \times 10^{-10}.$$

The finite- λ response satisfied

$$F_E(\lambda) \approx F_E(-\lambda)$$

for transverse perturbations, with median same-target evenness residual of approximately

$$7.5 \times 10^{-15},$$

and maximum residual below approximately

$$10^{-12}.$$

This established a highly accurate transverse even-order structure:

$$\boxed{F_E^{(T)}(\lambda) = \lambda^2 K_2 + \lambda^4 K_4 + \lambda^6 K_6 + \cdots} \quad (27)$$

12 Same-target symmetry and the origin of the even-order rule

The Γ symmetry connects $+E$ to $-E$, but it does not by itself explain why $F_E(\lambda)$ should be even in λ at the same target energy.

A complete Pauli-string search over the 8-qubit Pauli group,

$$4^8 = 65536$$

strings, was therefore performed for operators S satisfying

$$SH_0S^\dagger = H_0, \tag{28}$$

$$SV_TS^\dagger = -V_T. \tag{29}$$

Two strong candidates were found:

$$\boxed{S_1 = \text{YIIYXZY}, \quad S_2 = \text{XZZYXZI}.} \tag{30}$$

For both candidates:

- the Hamiltonian commutation relation held with residual $\sim 3.7 \times 10^{-15}$;
- the transverse sign flip was exact to machine precision;
- the target projectors were preserved with residuals of order 10^{-12} ;
- the finite- λ same-target covariance relation

$$S_P^\dagger F_E(\lambda) S_P = F_E(-\lambda)$$

held with maximum residual approximately 1.3×10^{-12} .

Thus the transverse even-order structure is not merely a numerical accident. It is controlled by a same-target symmetry.

Conceptually, Phase 3.2 therefore contains two distinct symmetry layers:

$$\Gamma : E \leftrightarrow -E, \tag{31}$$

$$S : \lambda \leftrightarrow -\lambda \quad \text{at fixed } E. \tag{32}$$

The first organizes the global spectral backbone; the second constrains the perturbative order structure of the transverse response.

13 Why the dephasing family behaves differently

The same-target symmetry operators S_1 and S_2 do not transform the dephasing perturbation as a simple global $\pm V_D$ parity.

For example, the dense validation produced residuals of order unity when testing

$$SV_DS^\dagger = \pm V_D.$$

Therefore, the even-order selection rule is family specific. The transverse family is symmetry protected under $\lambda \mapsto -\lambda$, whereas dephasing is not.

This cleanly explains the observed contrast:

$$\text{dephasing: } K_3 \neq 0, K_5 \neq 0 \text{ generically,} \quad (33)$$

$$\text{transverse: } K_3 \approx 0, K_5 \approx 0, \quad (34)$$

within the frozen reference model.

14 A unified mechanism for hotspot organization

The Phase 3.2 results can be organized into a single mechanistic chain.

14.1 Step 1: global spectral organization

The chiral-like operator Γ generates the exact $\pm E$ backbone:

$$\Gamma H_0 \Gamma^\dagger = -H_0.$$

14.2 Step 2: degeneracy-closed virtual pathways

For a chosen target eigenspace,

$$T_G = g_\eta(E - E_G) P V P_G V P$$

describes the effective contribution from each complete Q -space energy sector.

14.3 Step 3: coherent summation

The total effective response operator is

$$\Sigma_E = \sum_G T_G.$$

14.4 Step 4: concentration plus interference

The normalized coherence factor obeys

$$C_E^2 = S_E + A_E,$$

with S_E describing pathway concentration and A_E describing interference.

14.5 Step 5: effective response

The leading effective perturbative correction satisfies

$$\left\| \Delta H_{\text{eff}}^{(2)} \right\|_F = \lambda^2 C_E L_{1,E}.$$

Thus constructive virtual-path coherence directly enhances the local effective response.

14.6 Step 6: family-dependent order selection

For the transverse family, same-target symmetries S_1 and S_2 impose

$$SH_0S^\dagger = H_0, \quad SV_TS^\dagger = -V_T,$$

leading to a response constrained under $\lambda \mapsto -\lambda$ and hence to even perturbative order.

The complete qualitative picture is therefore

spectral symmetry $\rightarrow$ virtual pathways $\rightarrow$ coherence/interference $\rightarrow$ effective local response $\rightarrow$ hotspot organization

(35)

15 What has been established exactly, numerically, and statistically

To avoid overstatement, the conclusions should be classified carefully.

15.1 Exact or algebraic

The following relations are exact within the defined model:

$$\Sigma_E = \sum_G T_G, \tag{36}$$

$$\|\Sigma_E\|_F^2 = \sum_G \|T_G\|_F^2 + 2 \sum_{G < K} \text{Re Tr}(T_G^\dagger T_K), \tag{37}$$

$$C_E^2 = S_E + A_E, \tag{38}$$

$$\|\Sigma_E\|_F = C_E L_{1,E}. \tag{39}$$

The operator conditions

$$\Gamma H_0 \Gamma^\dagger = -H_0$$

and

$$S_i H_0 S_i^\dagger = H_0, \quad S_i V_T S_i^\dagger = -V_T$$

were numerically validated to machine precision for the frozen reference model.

15.2 Numerically established for the frozen reference model

The following statements are supported at or near machine precision:

- exact $\pm E$ projector pairing under Γ ;
- agreement between grouped and direct resolvent forms of Σ_E ;
- near-vanishing transverse K_3 and K_5 ;

- same-target transverse finite- λ covariance under S_1 and S_2 ;
- even-order transverse perturbative response.

15.3 Statistical and finite-size

The link between C_E , A_E and hotspot prominence is empirical rather than an exact identity. It is supported by:

- 12-seed analyses;
- strict sign-randomization controls;
- seed-level permutation tests;
- multiple Hamiltonian term counts;
- finite-size tests from 7Q through 12Q.

The correct formulation is therefore:

Virtual-path coherence is a robust mechanistic correlate and effective-response factor associated with hotspot organization.

It is not yet correct to state that hotspot prominence is exactly equal to a known function of C_E .

16 Relation to projection-operator methods

The structure used here is closely related to the familiar elimination architecture of projection methods:

$$PVQ(E - QHQ)^{-1}QVP.$$

A useful comparison is with projection-operator techniques in open quantum systems, where one encounters kernels schematically of the form

$$\mathcal{P}\mathcal{L}(t)G(t, s)\mathcal{Q}\mathcal{L}(s)\mathcal{P}.$$

The structural analogy is the repeated

$$P \rightarrow Q \rightarrow P$$

or

$$\mathcal{P} \rightarrow \mathcal{Q} \rightarrow \mathcal{P}$$

elimination architecture.

However, the distinction must remain explicit:

- the P, Q objects used in Phase 3.2 are Hilbert-space eigenspace projectors;

- in standard open-system projection techniques, $\mathcal{P}, \mathcal{Q}$ are commonly Liouville-space superoperators.

Likewise, the symbol Σ_E used here should not be identified automatically with unrelated Σ objects in open-system formalisms.

17 Limitations

Several limitations remain.

17.1 Finite-size regime

The largest explicitly tested systems contain 12 qubits. The observed robustness is therefore a finite-size statement, not an asymptotic theorem.

17.2 Frozen reference symmetry

The explicit operators

$$\Gamma = \text{XIXXIIXX}, \quad S_1 = \text{YYIIYXZY}, \quad S_2 = \text{ZXZZYXZI}$$

were established for the frozen reference construction. A separate symmetry search is required before claiming identical operator strings for arbitrary Hamiltonian seeds.

17.3 Prominence law

The final map from effective response to the exact empirical hotspot prominence statistic remains statistical.

17.4 Model specificity

The conclusions are derived inside the specific random-Pauli Hamiltonian framework and perturbation families used in the Soft Spaces project. Transfer to other Hamiltonian ensembles requires independent testing.

18 Reproducibility and computational lineage

The Phase 3.2 mechanism was developed through the following principal code sequence:

Version	Purpose
v32.0–v32.2	Identification and validation of the global Γ symmetry and exact $\pm E$ projector structure.
v32.3–v32.7	Local mechanism probes; tests of Σ_E , PVQ , spectral and operator descriptors.

Version	Purpose
v32.8–v32.11	Termwise interference, coherence-factor construction, randomization controls, and strict confirmation.
v32.12–v32.14	Compact local law, analytic interference decomposition, and degeneracy-closed basis-invariant formulation.
v32.15–v32.16.1	Robustness over Hamiltonian term count and 7Q–12Q system size.
v32.17–v32.20	Perturbative response tests, second-order checks, cross-seed order study, and operator-geometry alternatives.
v32.21	Exact coherence decomposition $C_E^2 = S_E + A_E$.
v32.22	Cross-Q coherence-decomposition confirmation.
v32.23	Analytic effective-response law $\ \Delta H_{\text{eff}}^{(2)}\ _F = \lambda^2 C_E L_{1,E}$.
v32.24	Direct perturbative-order decomposition into K_2, K_3, K_4 .
v32.25	Same-target evenness and explicit K_3, K_5 suppression for transverse response.
v32.26	Odd-order cancellation diagnostics.
v32.27	Exhaustive Pauli search revealing same-target symmetries S_1, S_2 .

19 Conclusions

Phase 3.2 began with a local-mechanism question and ends with a layered operator explanation.

The principal conclusions are:

1. A global chiral-like symmetry produces an exact $\pm E$ spectral backbone.
2. Local hotspot organization cannot be explained by raw P – Q coupling strength alone.
3. The relevant local structure is the coherent sum of degeneracy-closed virtual-pathway operators,

$$\Sigma_E = \sum_G T_G.$$

4. The normalized coherence factor

$$C_E = \frac{\|\Sigma_E\|_F}{\sum_G \|T_G\|_F}$$

is the most robust compact descriptor identified in this phase.

5. It obeys the exact decomposition

$$C_E^2 = S_E + A_E,$$

which separates pathway concentration from virtual-path interference.

6. C_E has a direct perturbative meaning:

$$\|\Delta H_{\text{eff}}^{(2)}\|_F = \lambda^2 C_E L_{1,E}.$$

7. The transverse perturbation family possesses a same-target symmetry that flips $\lambda \mapsto -\lambda$, producing an even-order response and suppressing odd coefficients such as K_3 and K_5 .

8. Two explicit Pauli-string generators of this same-target symmetry were found for the frozen reference model:

$$S_1 = \text{YIIYXZY}, \quad S_2 = \text{XZZYXZI}.$$

9. The virtual-path coherence mechanism remains positively associated with hotspot prominence across the investigated 7Q–12Q finite-size range.

The resulting mechanistic picture is compact:

global symmetry $\rightarrow$ degeneracy-closed virtual paths $\rightarrow$ interference/coherence $\rightarrow$ effective response $\rightarrow$ local ho

This provides a natural stopping point for Phase 3.2. Further work should focus on formal symmetry-algebra closure, generalization beyond the frozen reference Hamiltonian, and integration of the Phase 3.2 mechanism into the broader Soft Spaces theoretical framework.

Acknowledgement of status

This document is intended as a detailed technical synthesis of the completed Phase 3.2 numerical and analytical work. It distinguishes exact identities, machine-precision operator validations, and empirical finite-size/statistical relations. No asymptotic theorem is claimed.

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